Yehya Medhat

Electrical Engineer

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OBJECTIVE

Recent Electrical Engineering graduate eager to apply strong technical and problem-solving skills in a dynamic, results-driven engineering environment. Proactive and adaptable team player with a strong interest in control systems, automation, industrial robotics, and electrical engineering. Looking forward to contributing to innovative engineering projects while gaining valuable hands-on experience, expanding practical expertise, and developing within a professional industrial environment.

EDUCATION

Cairo University 2019 – 2024

B.Sc. in Electrical Engineering

- Graduated **first in class** with a **GPA of 3.77/4**.
- Graduation Project: IoT Applications in Smart Ambulances, grade: **A+**.
- Awarded **third place** in the university's graduation project exhibition.

College des Freres Bab El-Louk

Secondary Education

WORK EXPERIENCE

Siemens Energy Jul 2024 – Sep 2024

Electrical Engineer Intern

- Gained hands-on exposure to industrial motors, motor components, tooling, and maintenance processes at the Siemens service center.
- Worked with the robotics team on programming and controlling an industrial robotic arm used for coating applications.
- Participated in industrial site visits and technical and soft-skills training, gaining broader exposure to industrial engineering practices.

Skarpt Jul 2024 – Aug 2024

IoT Engineer Intern

- Programmed and configured home automation sensors and control devices, gaining practical experience with automation hardware and software systems.

Dar El-Handasah Jul 2023 – Aug 2023

Electrical Engineer Intern

- Applied electrical power distribution design principles, including load estimation, cable sizing, voltage-drop considerations, and transformer and generator selection.
- Developed building electrical designs, including lighting and power socket layouts using DIALux and AutoCAD.
- Completed and presented a full electrical design project applying the concepts learned throughout the internship.

National Air Navigation Services Company Aug 2022 – Sep 2022

Electrical Engineer Intern

COURSES

KNX Basics Dec 2024 – Jan 2025
KNX Association

Intelligent Control Oct 2023 – Dec 2023
Cairo University

Computer Control Oct 2023 – Dec 2023
Cairo University

Classic Control and PLCs Aug 2023 – Sep 2023
HA Consulting Group

Embedded Systems Jul 2023 – Oct 2023
IMT

PROJECTS

Inverted Pendulum Fuzzy Logic Controller
Control Systems

- Modeled nonlinear and linearized inverted-pendulum dynamics and implemented a fuzzy logic controller in Simulink for stabilization and disturbance recovery.

DC Motor System Identification
Control Systems

- Identified a second-order DC motor model from experimental data using MATLAB's System Identification toolbox.
- Designed and discretized a lead compensator and implemented the resulting difference equation on Arduino for closed-loop position control.

Self-Balancing Robot
Control Systems & Robotics Project

- Developed a self-balancing robot using sensor feedback and closed-loop PID control to maintain platform stability.

SKILLS

Control & Automation: PLC Programming, Control Design, Systems Integration, PID Tuning, Sensor Interfacing, Fuzzy Logic Design, PCB Design, KNX Programming

Communication Protocols: UART, SPI, I2C, CAN, Modbus

Programming: C/C++, Python

Software: Microsoft Office suite, TIA Portal, MATLAB/Simulink, AutoCAD

Soft Skills: Communication, Presentation, Critical Thinking, Problem Solving, Fast Learner, Self-Motivated

Languages: Arabic, English, French

EXTRACURRICULAR ACTIVITIES

Youth Leading Youth (YLY)
Operations Member – Volunteer

- Supported project coordination and event planning activities.
- Collaborated with team members to organize and execute events.
- Gained practical experience in teamwork, communication, and coordination.